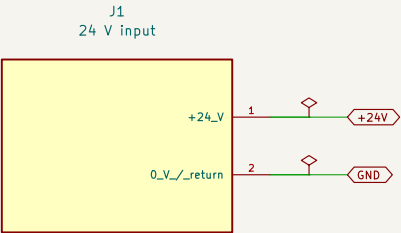
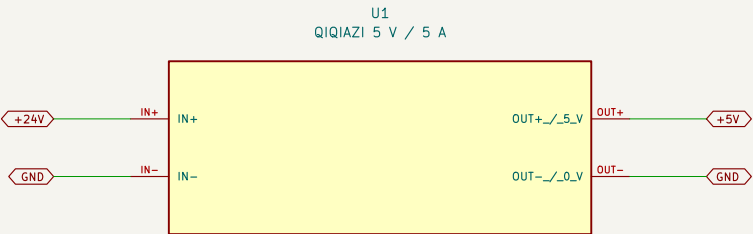


CONTROLLER AND POWER

A



B



Wire converter by IN+ / IN- / OUT+ / OUT- markings.
24 V also feeds all three motor drivers.
Common 0 V; separate power branches to each driver.
5 A is the converter rating, not the Nucleo input rating.

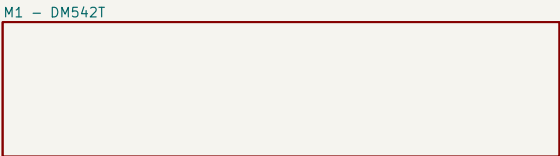
C

NUCLEO POWER CONFIGURATION
JP2: fit EXT jumper on pins 5–6.
Buck +5 V -> CN11 pin 6; ground -> CN11 pin 8.
CN8 pin 7 supplies 3.3 V to interface/storage.
5V_EXT input: 4.75–5.25 V, 500 mA maximum.
Power EXT first, then connect ST–Link USB CN1.
Do not connect 24 V to Nucleo VIN or logic pins.

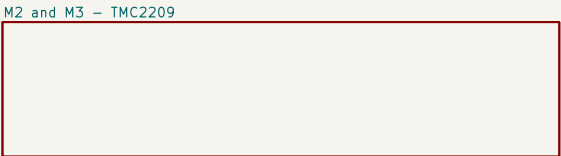
D

BOARD ROUTING (UM2407 table 14)
PE9 -> CN10.4: SB28 ON, SB70 OFF.
QSPI: SB59/SB63 ON; SB61/SB69 OFF.
PE2 -> CN10.25: SB67 OFF.
Only used external contacts are drawn.
Ethernet, ST–Link, LEDs and button stay onboard.

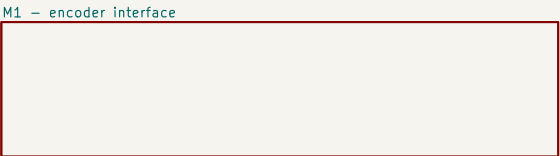
E



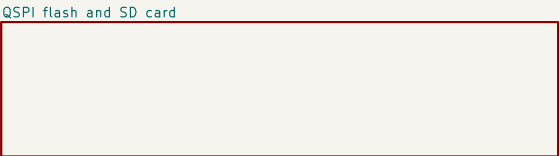
Date: sheets/M1_DM542T.kicad_sch



Date: sheets/M2_M3_TMC2209.kicad_sch

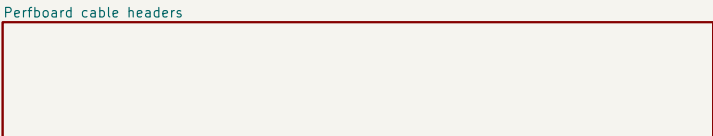


Date: sheets/M1_Encoder.kicad_sch



Date: sheets/Storage.kicad_sch

F



Date: sheets/Perfboard_Connectors.kicad_sch

Shared names connect across sheets. STEP signals use hardware timers.
A1 contacts use CNxx.pin numbering. J101–J107 are perfboard headers.
External modules are excluded from board; interface parts have linked footprints.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /

File: ProtoMate.kicad_sch

Title: ProtoMate SCARA – Controller and power

Size: A3

Date: 2026–10–02

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 1/6

A

B

C

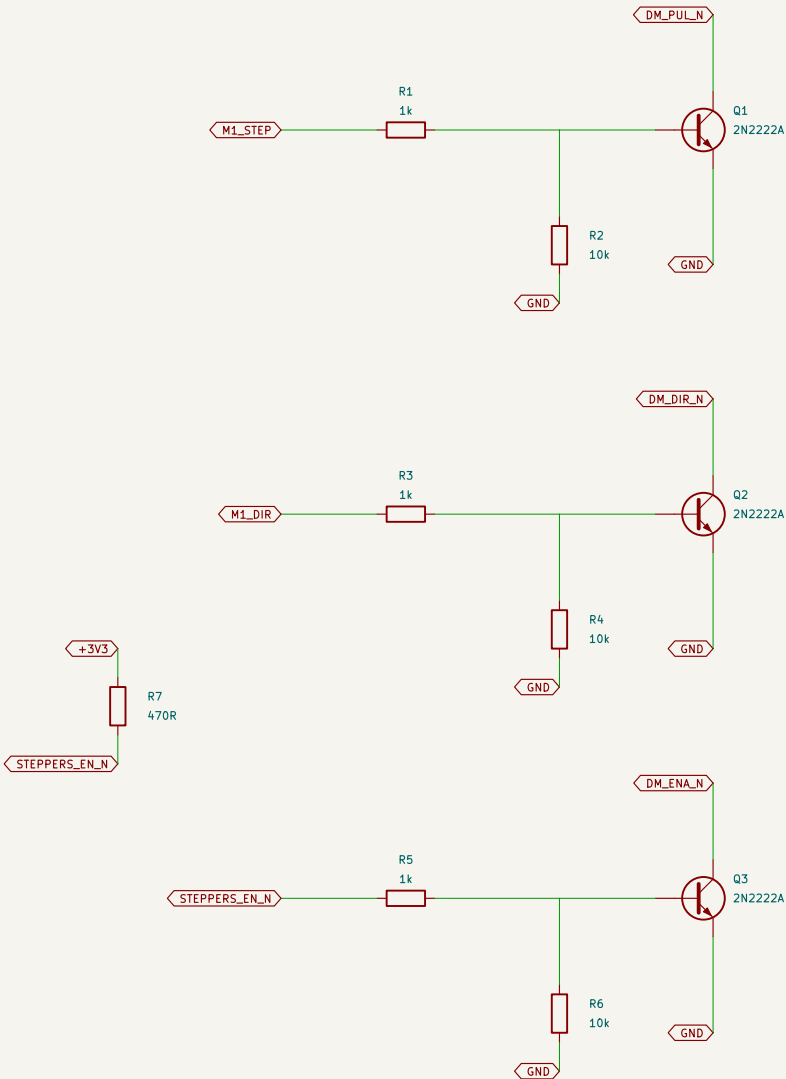
D

E

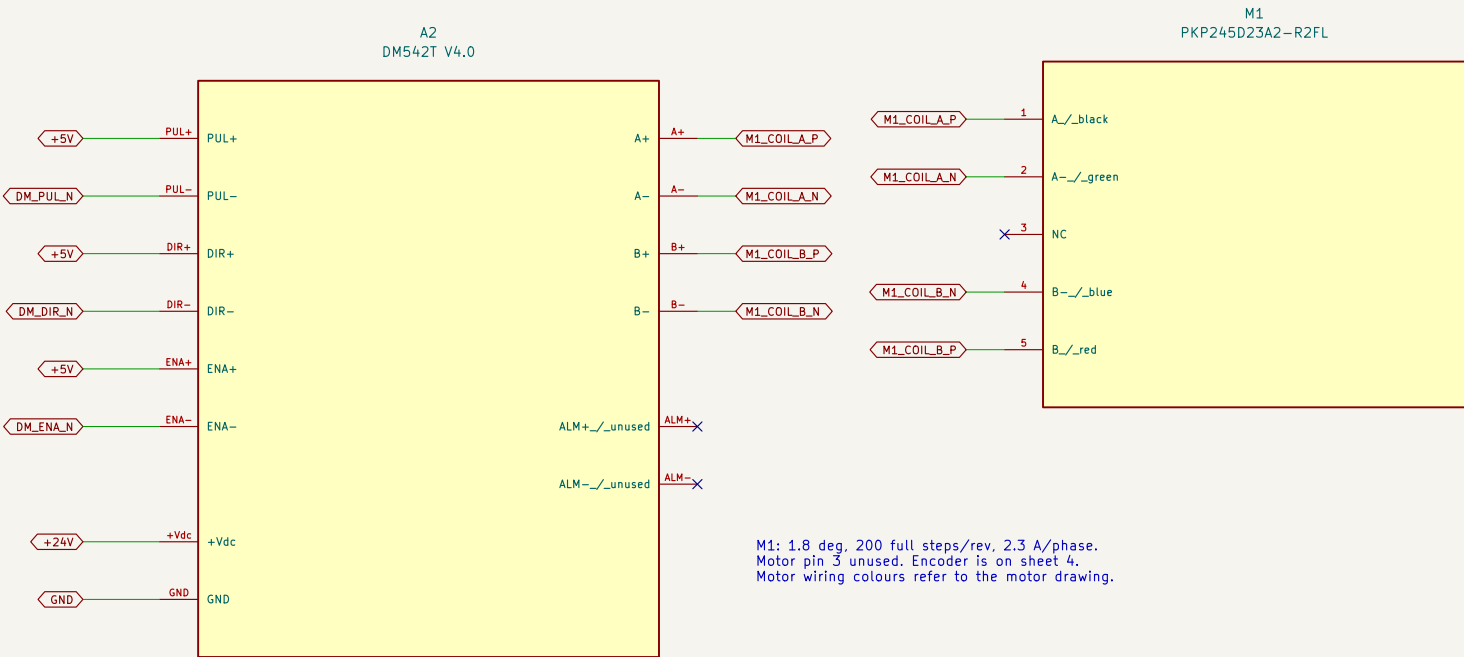
F

M1 – DM542T

3.3 V MCU -> 2N2222A current sinks -> 5 V optoisolated inputs



SHARED ENABLE AND TIMING
PF3 low: Q3 off, DM542T enabled, both TMC EN inputs low.
PF3 high: Q3 on, DM542T disabled, both TMC EN inputs high.
R7 pulls PF3 high during MCU reset; about 7 mA sink when enabled.
After enable, wait >=200 ms. DIR setup >=5 us; STEP high/low >=2.5 us.



DM542T SETTINGS / INITIAL COMMISSIONING
S2 signal selector: 5 V.
1/16 step: SW5 ON, SW6 ON, SW7 OFF, SW8 ON.
3200 STEP pulses per motor revolution.
Conservative initial current: 2.03 A RMS / 2.84 A peak
SW1 ON, SW2 ON, SW3 OFF; SW4 OFF (idle reduction).
Check the actual V4.0 label before setting switches.

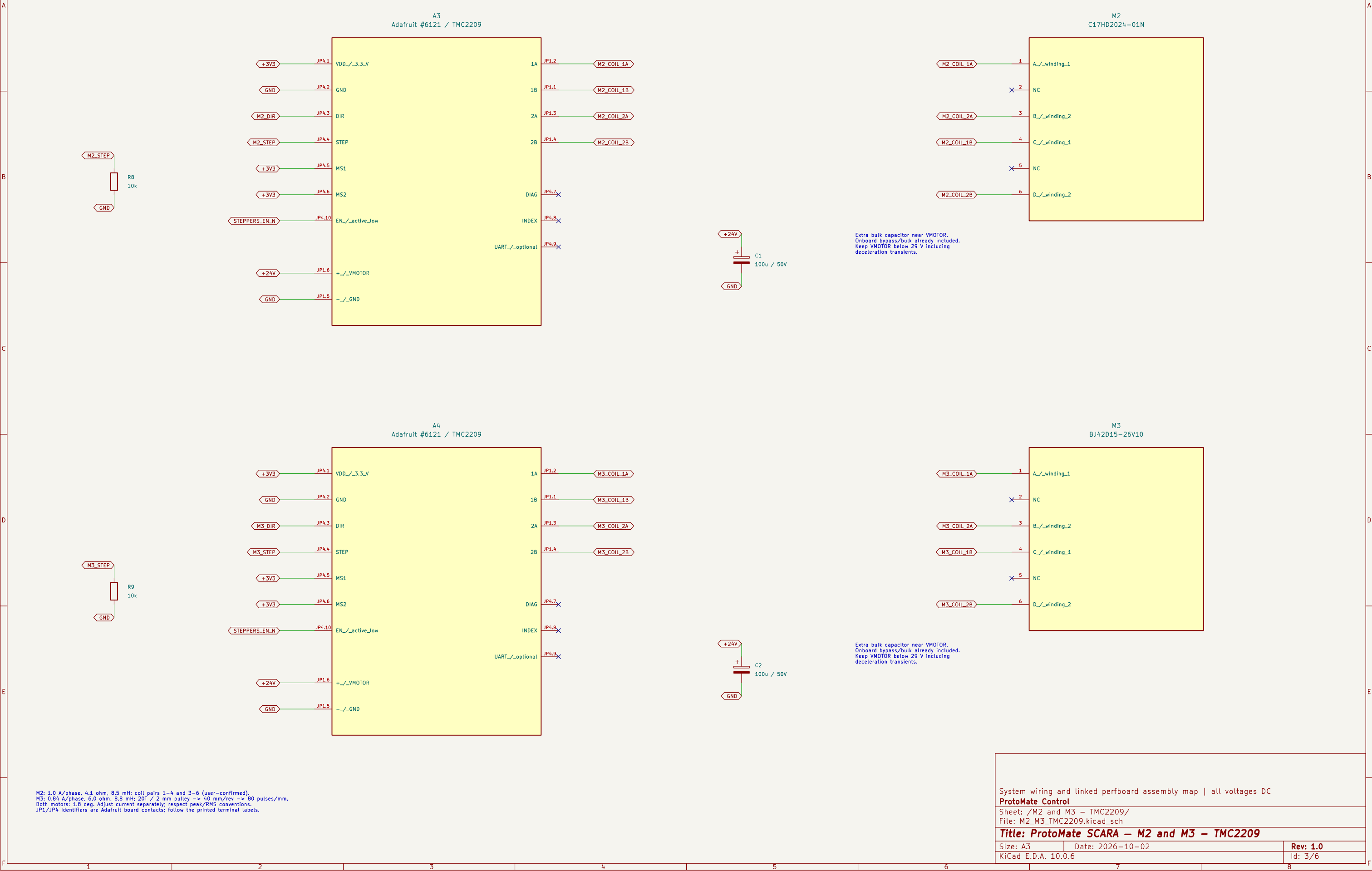
Q1–Q3 show functional B/C/E pins, not package lead numbers.
Verify your 2N2222A manufacturer/package before assembly.
Fit each 10k between base and emitter. Keep PUL+/PUL- paired;
separate signal cables from motor power cables.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control		
Sheet: /M1 – DM542T/ File: M1_DM542T.kicad_sch		
Title: ProtoMate SCARA – M1 – DM542T		
Size: A3	Date: 2026–10–02	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 2/6

M2 AND M3 – TMC2209

MS1 = MS2 = VDD (3.3 V): 1/16 step, 3200 pulses/rev. No MCU pins needed.
SPRD jumper open: factory StealthChop mode. Current set with each onboard potentiometer.

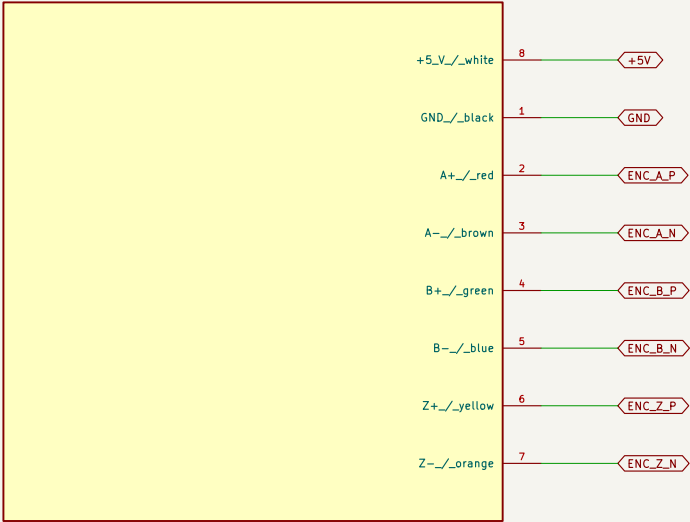


M1 – ENCODER / HC126 + NPN LEVEL SHIFTERS

U2 input pairs are deliberately reversed.
Q4–Q6 cancel that inversion before U3.
Encoder connector pinout stays unchanged.
Never wire U2 outputs directly to U3.

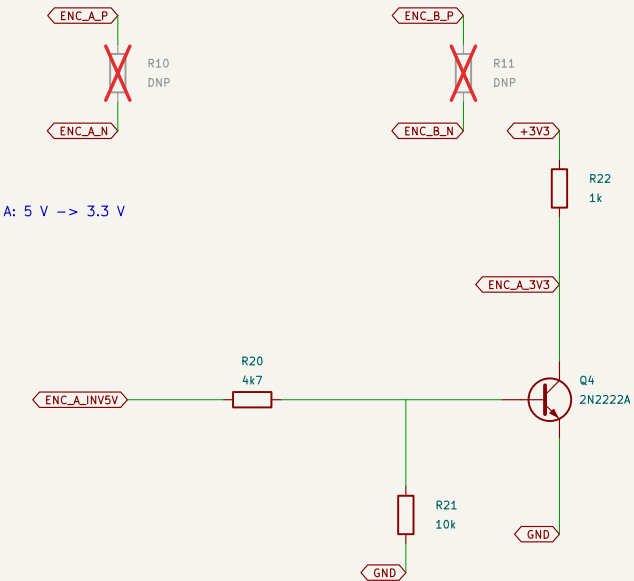
U2: 5 V. Q4–Q6 translate to 3.3 V. U3: 3.3 V ONLY.
U3 OE1/OE2/OE3 HIGH; unused OE4 and A4 LOW.
U2 unused receiver inputs may float (fail–safe).
100 nF at each IC; 1 uF nearby.

J2
M1 encoder / 8–pin



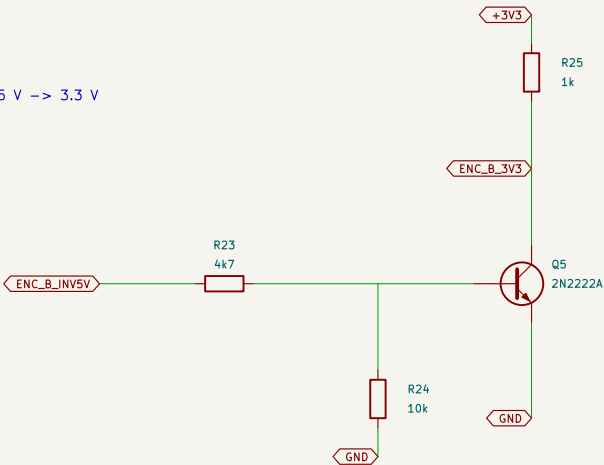
400 P/R -> TIM3 x4 = 1600 counts/rev; Z = once/rev.
Cable colours / connector orientation; motor manual.
Shielded twisted pairs; chassis bond at cable entry.

R10–R12: OPTIONAL TERMINATION, DNP.
Choose for cable and 20 mA encoder output limit.
A+/A-, B+/B-, Z+/Z-: twisted pairs.

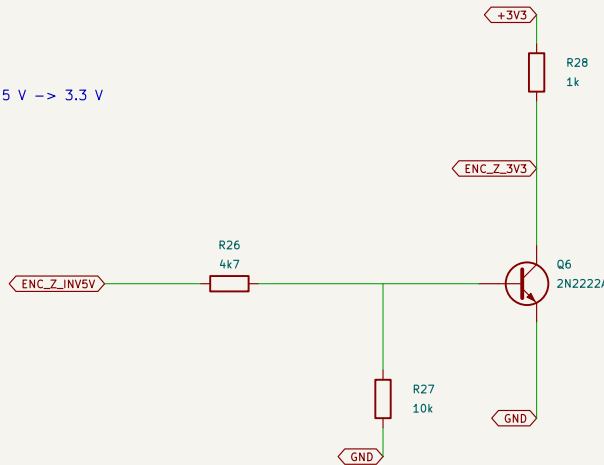


A: 5 V -> 3.3 V

B: 5 V -> 3.3 V



Z: 5 V -> 3.3 V



A/B -> PB4/PB5; Z -> PB6 rising–edge EXTI. All original signal polarities preserved.
Q4–Q6: Diotec 2N2222A E=1/B=2/C=3. Keep collector–to–U3 wires short.
Check edge shape and transistor turn–off delay at the intended encoder speed.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M1 – encoder interface/
File: M1_Encoder.kicad_sch

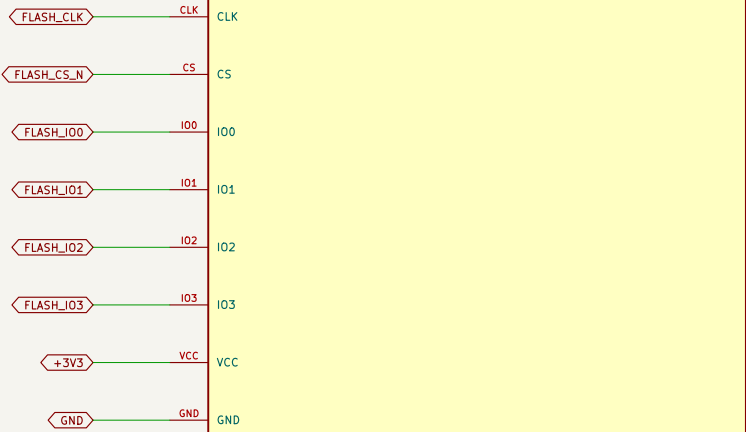
Title: ProtoMate SCARA – M1 – encoder interface

Size: A3 Date: 2026–10–03 Rev: 2.0
KiCad E.D.A. 10.0.6 Id: 4/6

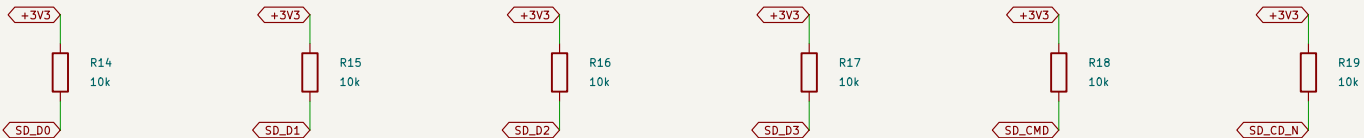
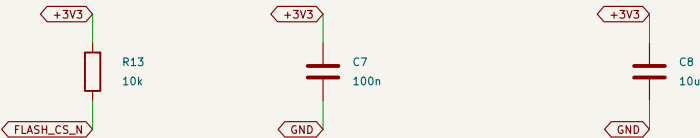
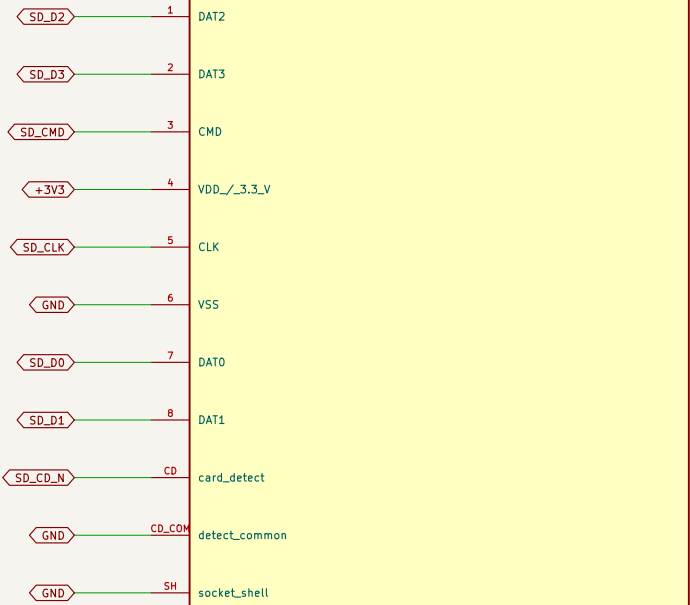
QSPI FLASH AND SD CARD

SD pins 1–8 use standard card–contact numbering.
CD switch closes to GND when inserted (advisory input).
Pull–ups on CMD/D0–D3/CD; no pull–up on CLK.
Existing equivalent module pull–ups/caps need not be duplicated.

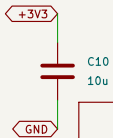
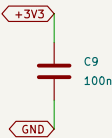
A5
W25Q128 / 3.3 V QSPI



J3
microSD / 4–bit socket module



3.3 V ONLY
Use a W25Q128 module exposing IO0–IO3.
IO2/WP and IO3/HOLD are data in quad mode;
do not hard–wire them to supply or ground.
Module contacts are identified by signal name.
C7/C8 and R13 may already exist on the module;
fit them if absent. Keep the QSPI harness short.



System wiring and linked perfbord assembly map | all voltages DC

ProtoMate Control

Sheet: /QSPI flash and SD card/

File: Storage.kicad_sch

Title: ProtoMate SCARA – QSPI flash and SD card

Size: A3

Date: 2026–10–02

Rev: 1.0

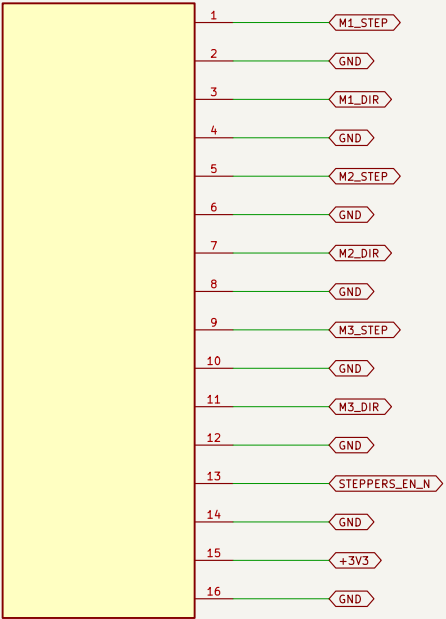
KiCad E.D.A. 10.0.6

Id: 5/6

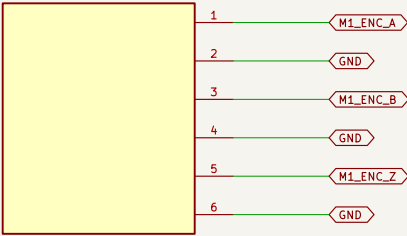
PERFBOARD CABLE HEADERS

Pin numbers match the layout and Harness.csv. All headers are 2.54 mm pitch; every drawn contact is on the perfboard.

J101
Nucleo outputs / 2x08



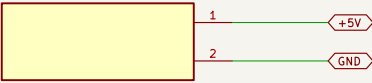
J102
Nucleo encoder / 2x03



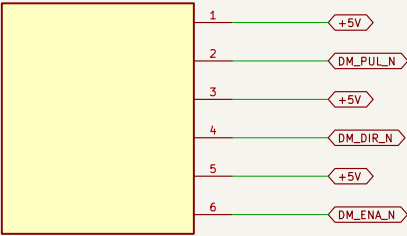
J101/J102: paired even pins are GND.
J101 pin 15 receives Nucleo 3.3 V.
J103: regulated 5 V from buck only.
24 V and motor currents stay off this board.

J104: 1 PUL+, 2 PUL-, 3 DIR+, 4 DIR-, 5 ENA+, 6 ENA-.
J105/J106: 1 VDD, 2 GND, 3 DIR, 4 STEP, 5 MS1, 6 MS2,
7 DIAG (NC), 8 INDEX (NC), 9 UART (NC), 10 EN.
J107 follows the numbered J2 encoder connector.

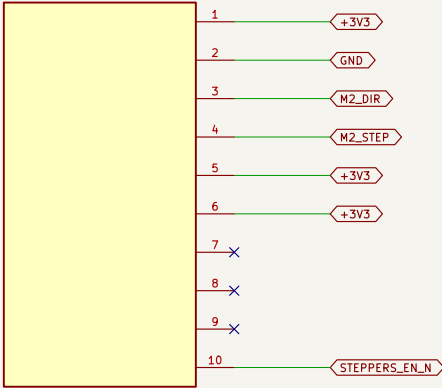
J103
Buck 5V / 1x02



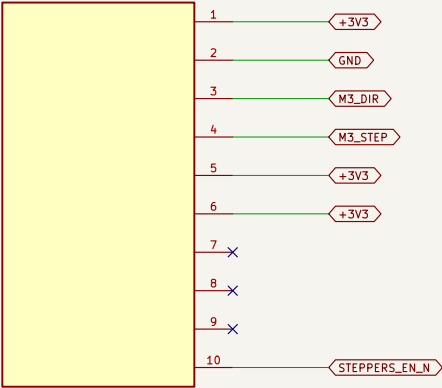
J104
DM542T / 1x06



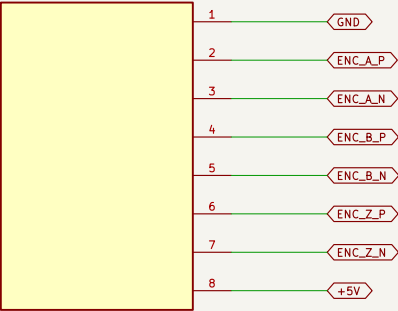
J105
M2 TMC2209 / 1x10



J106
M3 TMC2209 / 1x10



J107
M1 encoder / 1x08



Read Harness.csv for destination contacts (CNxx.pin / driver labels).
External module symbols elsewhere show the far end of these cables. Matching names represent the same conductor.

All J101–J107 headers mount on the perfboard; signals connect by global names

ProtoMate Control

Sheet: /Perfboard cable headers/

File: Perfboard_Connectors.kicad_sch

Title: ProtoMate SCARA – Perfboard cable headers

Size: A3

Date: 2026–10–03

Rev: 3.0

KiCad E.D.A. 10.0.6

Id: 6/6